

Cross-View Visual Uncertainty Routing for Training-Free Multimodal Confidence

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Abstract

Multimodal language models need calibrated uncertainty estimates when visual evidence is incomplete, ambiguous, or text-heavy, but many practical deployments cannot retrain the model or read internal logits. We study Cross-View Visual Uncertainty Routing (CVUR), a training-free protocol that asks whether a candidate answer remains stable across prompt, crop/zoom, caption-style evidence, and OCR-oriented views of the same image-question pair. We evaluate CVUR with Qwen/Qwen2.5-VL-7B-Instruct under deterministic decoding, a 64-token answer budget, and a bounded public slice spanning POPE, HallusionBench, MM-Bench English dev, and TextVQA validation. The slice is designed to stress hallucination, visual conflict, broad reasoning, and OCR-sensitive grounding. On 240 proposed-method rows, CVUR reaches 0.720833 pooled accuracy, compared with 0.716667 for the strongest baseline, but the paired sign test against that baseline is non-significant ($p = 1.000000$). The stronger evidence is selective-prediction behavior: at the recorded threshold, CVUR keeps 0.937500 coverage with 0.257778 selective risk and a 0.062500 route/abstain rate, with ECE-10 of 0.194792 and Brier score of 0.233073. We therefore frame CVUR not as a broad accuracy winner, but as a lightweight confidence interface for deciding when a model answer is stable enough to trust and when it should be routed or abstained.

1 Introduction

Multimodal question answering systems increasingly operate in settings where an answer is useful only if the system can also say when its visual evidence is unreliable. A model may answer a visual question confidently while the relevant object is cropped, the requested text is small, or the premise is contradicted by another view of the image. Existing work on calibration shows that neural confidence can be badly misaligned with correctness

(Guo et al., 2017), and selective prediction formalizes the option to reject low-confidence cases (Geifman and El-Yaniv, 2019). In multimodal language models, the same issue is sharper because uncertainty can arise from visual grounding, OCR, object hallucination, and language priors at once (Chen et al., 2024; Du et al., 2026; Tran et al., 2026).

Benchmark studies make these failure sources concrete: POPE targets object hallucination, HallusionBench targets entangled visual illusion and language conflict, and OCR-centered VQA work shows that small or stylized text can change the answer even when the scene appears visually simple (Li et al., 2023; Guan et al., 2023; He et al., 2025).

Recent multimodal uncertainty methods provide important evidence that richer uncertainty features can outperform simple confidence scores. Incoherence-adjusted semantic volume measures the dispersion and incoherence of sampled multimodal responses (Lau et al., 2026), uncertainty-aware optimization adapts model preferences to uncertainty (Zhang et al., 2026a), and uncertainty-aware VQA models study selective prediction directly (Wieczorek et al., 2025). These directions motivate a complementary question: how far can a system go without training, internal logits, or white-box access? This question matters for deployed hosted models, local open-weight models without calibration data, and benchmark audits where the model should be treated as a black-box answerer.

We propose Cross-View Visual Uncertainty Routing, abbreviated CVUR. The protocol asks the same image-question pair through four independent views: a multi-prompt answer view, a crop/zoom view, a caption-style evidence view, and an OCR-oriented view. Rather than treating any one view as an oracle, CVUR converts their agreement into a confidence score and applies a thresholded decision: trust the answer, or route/abstain. This paper studies whether such cross-view consistency gives a useful calibrated confidence signal for mul-

timodal LLM answers.

Our evidence supports a narrow claim. Across four public benchmark families, each represented by a 60-row slice, CVUR obtains a small pooled accuracy edge over the strongest baseline: 0.720833 versus 0.716667. That edge is not statistically meaningful in the paired comparison to the strongest baseline ($p = 1.000000$), and family-level results include ties and underperformance. The more stable finding is that CVUR supplies a selective-prediction interface: at its recorded threshold it abstains or routes 6.25% of examples, retains 93.75% coverage, and reports lower ECE-10 than the always-trust and verbal self-confidence baselines.

This paper makes three contributions. First, it defines a training-free visual uncertainty protocol whose units are easy to inspect: prompt consistency, crop/zoom consistency, caption-VQA consistency, and OCR agreement. Second, it evaluates the protocol on a completed four-family benchmark slice with baselines that represent always trusting the answer, verbal self-confidence, temperature-mapped confidence, and a black-box semantic-volume approximation. Third, it reports calibration, selective prediction, paired tests, ablations, and failure categories with claim downgrades where the evidence is weak. The central conclusion is practical rather than sweeping: cross-view consistency is a plausible confidence interface, but the current evidence does not justify claiming broad accuracy superiority.

2 Related Work

Calibration and uncertainty for multimodal models. Calibration studies ask whether confidence values are meaningful probabilities of correctness. The classic finding that modern neural networks are often miscalibrated (Guo et al., 2017) extends naturally to multimodal settings, where the answer depends on both visual grounding and language priors. Recent MLLM calibration work evaluates empirical miscalibration across tasks and model families (Chen et al., 2024; Du et al., 2026), while adaptive confidence refinement asks when an audio-visual system should answer (Tran et al., 2026). CVUR follows this line but uses agreement across black-box views rather than a learned calibration head.

Selective prediction and routing. Selective prediction introduces a reject option for examples

whose expected risk is too high (Geifman and El-Yaniv, 2019). Recent LLM work argues that entropy alone can be insufficient for safe selective prediction (Phillips et al., 2026), and controlled-risk deployment work studies risk-aware operating points for large language models (Zellinger and Thomson, 2024). The CVUR threshold is a selective-prediction decision, but the paper deliberately evaluates route/abstain behavior rather than downstream tool-answer utility.

Consistency and semantic uncertainty.

Semantic-volume methods estimate uncertainty through response diversity and semantic disagreement (Lau et al., 2026). Preference-optimization and variational VQA approaches incorporate uncertainty into training or model design (Zhang et al., 2026a; Wiczeorek et al., 2025). CVUR is closest in spirit to black-box consistency scoring: it asks whether independently prompted evidence views support the same answer. Its distinction is operational rather than architectural: the protocol uses no gradient updates, no internal logits, and no learned verifier.

Multimodal hallucination, VQA, and OCR-sensitive evaluation.

Visual hallucination benchmarks and analyses show that object presence, visual illusion, and text cues can expose failures that aggregate accuracy hides (Guan et al., 2023; Li et al., 2023; Wu et al., 2024; Yan et al., 2024). Hallucination repair and mitigation work further shows that external checks or self-reward phases can reduce some errors (Yin et al., 2023; Zhang et al., 2026b). OCR-heavy tasks require a different type of grounding because the correct answer may depend on small or stylized text (He et al., 2025). These strands motivate CVUR’s four signals: prompt, crop/zoom, caption evidence, and OCR agreement.

3 CVUR: Consistency Signals for Visual Uncertainty

CVUR is a wrapper around an existing multimodal answerer. Given an image x , a question q , and a candidate answer a , the wrapper returns a scalar confidence score $s(x, q, a)$ and a decision $d \in \{\text{trust}, \text{route/abstain}\}$. The answerer is not retrained. The only intervention is to query or score the same instance through multiple views that should agree when the answer is visually grounded.

The first signal is multi-prompt consistency. The

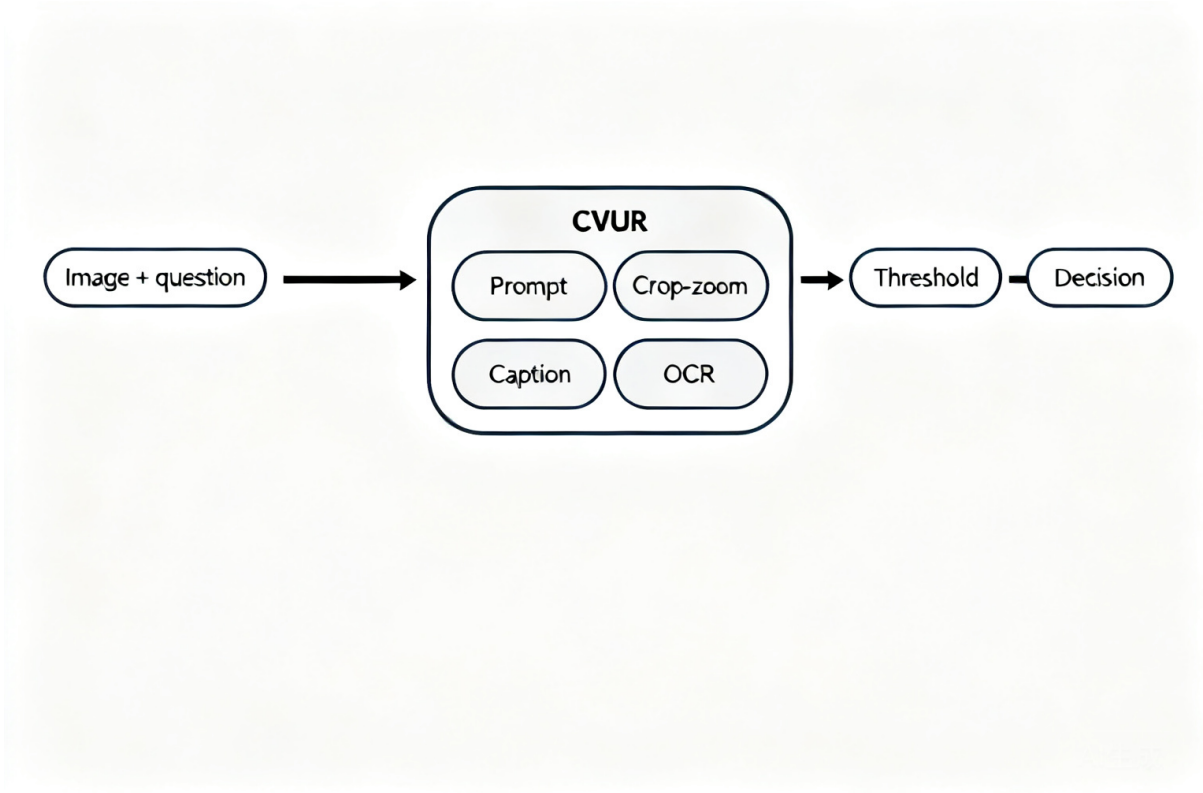


Figure 1: **CVUR converts four visual consistency checks into a thresholded trust decision.** The method overview shows the decision path: image and question enter CVUR, view checks compare prompt, crop/zoom, caption, and OCR agreement, and a threshold routes the answer to trust or abstention. The numeric evidence is not encoded in the raster; Tables 2 and 3 report the bounded four-family evaluation, including the 0.720833 versus 0.716667 pooled accuracy comparison and non-significant strongest-baseline paired test.

answerer is asked semantically equivalent variants of the question, and CVUR measures whether the resulting answers remain compatible with the candidate. This signal captures sensitivity to phrasing without assuming that one prompt is privileged. The second signal is crop/zoom consistency. The model is asked about a focused or altered visual view when the benchmark image and question make such a view meaningful. This signal tests whether the answer depends on global image context alone or survives a closer visual check.

The third signal is caption-VQA consistency. A caption-style evidence prompt asks the model to describe relevant visual facts before or alongside the answer. Agreement between that evidence and the candidate answer is treated as support; contradiction or missing evidence lowers confidence. The fourth signal is OCR agreement. For text-sensitive images, an OCR-oriented view asks for visible text or text-grounded evidence. This signal is important because a visually plausible answer may still be unsupported when the relevant text is unreadable or absent.

The four scores are aggregated into a single confidence value. The experiments use a fixed aggregation and a pre-specified operating threshold recorded before test-set interpretation. The score is therefore best interpreted as an operational trust score, not as a calibrated probability learned from held-out labels. If the score exceeds the threshold, the system trusts the candidate answer; otherwise, it routes or abstains. Figure 1 intentionally keeps this decision topology simple, while the caption and tables carry the benchmark and calibration details.

Two boundaries are central to the method claim. First, CVUR is training-free but not cost-free: it spends additional view queries or scores to obtain the confidence signal. Second, route/abstain is measured as selective prediction in this paper. The experiment does not evaluate a downstream tool caller that produces a corrected final answer after routing. Consequently, the method claim is about reliability-aware decision support, not end-to-end tool-use improvement.

Table 1: **The four CVUR signals target different visual uncertainty sources before one thresholded trust decision.** The table describes the paper-facing mechanism; aggregate effects are measured in Tables 2 and 3.

Signal	What means	agreement	Failure tested	pressure
Prompt consistency	Rephrasings preserve the answer	preserve	Language-prior sensitivity	
Crop/zoom consistency	Focused visual evidence stays compatible	visual evidence	Missing or peripheral object evidence	
Caption-VQA consistency	Described facts support candidate	visual support	Unsupported visual rationalization	
OCR agreement	Visible text supports the answer when relevant	Text supports	Text-reading and small-token errors	

Signal construction. The protocol treats each consistency check as a noisy observation about the same candidate answer rather than as a separate answer generator. The multi-prompt view checks whether short rephrasings of the question preserve the answer semantics. The crop/zoom view checks whether the answer remains compatible when the visual field is focused on the region most likely to contain the evidence. The caption-evidence view asks for visual facts in a descriptive mode and then compares those facts with the candidate answer. The OCR-oriented view is activated for text-sensitive cases and asks whether visible text supports the answer. Each view can support, contradict, or fail to supply evidence for the candidate.

The aggregation rule is intentionally simple. A candidate that receives coherent support from more views receives a higher score, while contradiction or missing evidence lowers the score. This design makes the score auditable: if an answer is trusted incorrectly, the failure can be traced to which views agreed with the wrong answer. The design also avoids a hidden learned calibration layer, which is important for the training-free setting studied here. The cost is that CVUR cannot learn family-specific reliability weights from data; when a benchmark family stresses a signal that the current aggregation does not weight well, the fixed score can underperform a simpler baseline.

Threshold and operating point. The decision threshold is part of the recorded evaluation protocol. The main tables therefore report coverage, selective risk, and route/abstain rate at that operating

point rather than selecting a threshold after looking at test outcomes. This distinction matters because the same score can support different deployment policies. A human-review workflow might accept lower coverage to reduce trusted errors; an interactive assistant might keep higher coverage and use routing only for the most unstable answers. The present evidence evaluates the recorded operating point and uses threshold sweeps only to describe the coverage-risk curve.

What CVUR is not. The wrapper does not inspect hidden states, logits, or gradients. It does not train a verifier, learn a calibration map, or call an external tool to repair routed examples. It also does not claim that view agreement is equivalent to correctness. The central risk is correlated error: if the base model repeats the same mistaken object, text reading, or commonsense prior across views, consistency can make a wrong answer look stable. The failure analysis below is included to keep that risk visible in the main body rather than relegating it to a supplement.

Experimental setup.

Benchmarks. The evaluation uses four public multimodal benchmark families: POPE, HallusionBench, MMBench English dev, and TextVQA validation. Each family contributes 60 deterministic task identifiers sampled before scoring. This gives 240 proposed-method examples and matching baseline examples. These are real benchmark rows with ground-truth scoring, but they are not official full benchmark submissions or leaderboard-scale evaluations.

The four families were chosen to cover different reasons that visual confidence can be misleading. POPE stresses absent-object false trust. HallusionBench stresses language-visual conflict and illusion-like cases. MMBench English dev contributes broader multimodal reasoning examples and is the least favorable family for CVUR in several comparisons. TextVQA validation stresses OCR-sensitive questions, where a plausible visual answer can be wrong if the relevant text is not read. The slice therefore functions as a stress-test bundle for confidence behavior, not as a leaderboard substitute.

Evaluated model and baselines. All paper-facing conditions use Qwen/Qwen2.5-VL-7B-Instruct as the multimodal answerer with deterministic decoding at temperature 0.0 and a 64-token

answer budget. The baselines are Always Trust, Verbal Self-Confidence, Temperature-Mapped Self-Confidence, and Black-Box Semantic-Volume Umpire. Always Trust accepts every answer; verbal and temperature-mapped confidence use answer-side confidence proxies; the semantic-volume baseline approximates black-box answer variation.

The baselines are deliberately black-box because the motivating setting does not assume access to model internals. Always Trust represents deployments that accept every generated answer. Verbal Self-Confidence represents answer-side self-reporting. Temperature-Mapped Self-Confidence represents a simple monotonic confidence transformation. Black-Box Semantic-Volume Umpire represents a stronger consistency baseline that estimates answer dispersion without using the CVUR view taxonomy. This baseline is important for interpretation: CVUR only edges it by 0.004167 pooled accuracy in the paired table, so the paper does not treat semantic-volume-style consistency as solved or dominated.

The benchmark harness applies the same image-question tasks, answer budget, and deterministic decoding policy to every method. CVUR receives the base candidate answer and the four cross-view consistency observations, aggregates them into one trust score, and compares that score with the recorded operating threshold. A trusted case contributes the model answer under selective prediction; a routed or abstained case is counted as not trusted rather than repaired. Baselines receive the same task set and differ only in the confidence signal used for the trust decision.

Metrics. We report accuracy, ECE-10, Brier score, coverage at the recorded threshold, selective risk, and route/abstain rate. For paired binary outcomes, we report sign-test p -values and avoid significance claims when the test does not support them. For ablations, we compare CVUR with variants that remove or simplify individual signals. These ablations are diagnostic: several variants tie the full method on this slice, so they cannot establish that every signal is individually necessary.

Accuracy is included because readers need to know whether routing decisions are coupled to answer correctness, but it is not the only success criterion. ECE-10 and Brier score measure whether the confidence score behaves like a reliability signal. Coverage and selective risk measure the cost of ab-

stention or routing. Paired sign tests protect against over-reading small pooled deltas. The main result is therefore a bundle of claims: a small pooled accuracy edge, a clearer calibration advantage over the simple confidence baselines, and an operating point that routes a small share of cases while retaining most coverage.

Claim scope. The completed four-family evidence supports the selective-prediction result more strongly than it supports the pooled accuracy advantage. The paper therefore states the pooled edge numerically, but does not present it as a uniform or statistically significant win.

This scope also controls what is omitted. The manuscript does not claim state-of-the-art multimodal VQA accuracy, does not claim that CVUR improves every benchmark family, and does not claim that every named signal is necessary. Those statements would require larger official splits, additional model families, and stronger ablation evidence. The present paper instead asks a narrower question: whether a training-free cross-view score can be a useful trust signal on a completed multi-family slice.

4 Results

Table 2 is the central cross-benchmark matrix. CVUR has the highest pooled accuracy, but the accuracy difference is small and the family rows are mixed. The strongest baseline by pooled accuracy is Black-Box Semantic-Volume Umpire at 0.716667, only 0.004166 below CVUR. This is why the paper frames CVUR as a confidence and routing protocol rather than an accuracy replacement for stronger semantic-volume baselines.

The calibration results also show why the result is not reducible to accuracy. Always Trust has full coverage but a selective risk equal to its error rate. Temperature-mapped and verbal confidence route more examples but do not improve accuracy on the pooled slice. CVUR keeps 0.937500 coverage and a 0.062500 route/abstain rate, while reducing ECE-10 from 0.291667 for Always Trust and 0.310208 for Verbal Self-Confidence to 0.194792. The semantic-volume baseline has slightly lower selective risk at its threshold, so CVUR should not be described as dominating every selective metric.

Figure 2 visualizes the selective-prediction trade-off. The intended use of CVUR is not to reject as many examples as possible. Instead, it supplies a confidence score that can be tuned for a

Table 2: **CVUR reaches 0.720833 pooled accuracy versus 0.716667 for the strongest baseline, with mixed family-level behavior.** All rows use Qwen/Qwen2.5-VL-7B-Instruct, deterministic decoding at temperature 0.0, a 64-token answer budget, and 60 tasks per benchmark family.

Source family	Tasks	Backend/budget	Always	Verbal	Temp.	SemVol	CVUR
HallusionBench	60	Qwen2.5-VL-7B; temp. 0; 64 tokens	0.816667	0.833333	0.833333	0.800000	0.833333
MMBench English dev	60	Qwen2.5-VL-7B; temp. 0; 64 tokens	0.333333	0.366667	0.366667	0.366667	0.350000
POPE	60	Qwen2.5-VL-7B; temp. 0; 64 tokens	0.850000	0.833333	0.833333	0.850000	0.850000
TextVQA validation	60	Qwen2.5-VL-7B; temp. 0; 64 tokens	0.833333	0.800000	0.800000	0.850000	0.850000
Pooled slice	240	Qwen2.5-VL-7B; temp. 0; 64 tokens	0.708333	0.708333	0.708333	0.716667	0.720833

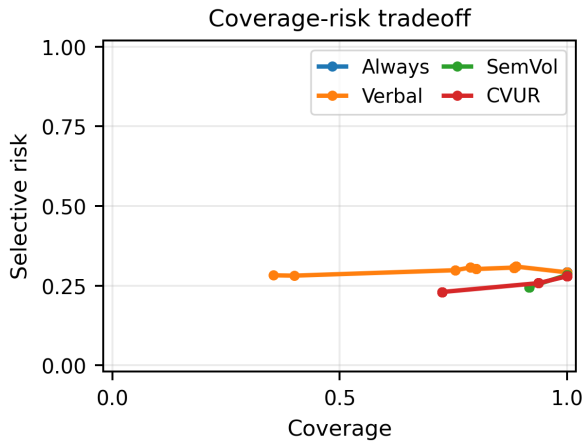


Figure 2: **CVUR retains 0.937500 coverage with 0.257778 selective risk at the recorded threshold.** The curve shows the trust-versus-route operating tradeoff.

deployment-specific risk target. At the recorded threshold, CVUR routes only a small fraction of examples, which limits how much selective risk can fall. A more aggressive threshold would change the coverage-risk tradeoff and should be selected using a validation policy rather than by inspecting test outcomes.

The paired tests in Table 3 are the main guardrail on interpretation. Most examples are ties because the evaluated answer is often unchanged across confidence methods; the methods differ primarily in the route/abstain decision and in a small number of answer-scoring cases. Against the strongest baseline, CVUR records two wins, one loss, and 237 ties, yielding $p = 1.000000$. The correct reading is a small pooled advantage with no statistical evidence for superiority over that baseline.

Calibration, ablation, and failure analysis. The calibration results support the use of CVUR as an

Table 3: **Diagnostics support cautious wording: strongest-baseline paired evidence is 2/1/237 with $p = 1.000000$, while removing caption-VQA reduces pooled accuracy by 0.020833.** The paired and ablation panels are diagnostic rather than proof of uniform superiority or component necessity.

Diagnostic	Effect	W/L/T	p
vs. Always Trust	+0.012500	5/2/233	0.453125
vs. Semantic-Volume Umpire	+0.004167	2/1/237	1.000000
minus caption-VQA	-0.020833	0/5/235	0.062500
minus OCR agreement	-0.008333	0/2/238	0.625000

operating-point interface. ECE-10 improves from 0.291667 for Always Trust and 0.310208 for Verbal Self-Confidence to 0.194792 for CVUR. The Brier score follows the same direction: 0.233073 for CVUR compared with 0.291667 for Always Trust and 0.334198 for Verbal Self-Confidence. Temperature-Mapped Self-Confidence is closer on ECE-10 at 0.212240, but has lower coverage and higher selective risk at the recorded threshold.

Figure 3 makes the family-level caveat visible. The four-family average hides important heterogeneity. HallusionBench and POPE are favorable or tied for CVUR in several comparisons, while MMBench English dev includes underperformance against confidence baselines and the semantic-volume baseline. TextVQA validation is favorable for some comparisons but not enough to turn the paired aggregate into a significant result.

The ablation rows in Table 3 weaken a tempting method story. Caption-VQA consistency is the most useful signal in the pooled ablation table, but exact-match-only, crop/zoom-removed, no-routing, and per-family-threshold variants tie CVUR on this evidence slice. The right conclusion is that cross-view evidence can be useful, not that every named

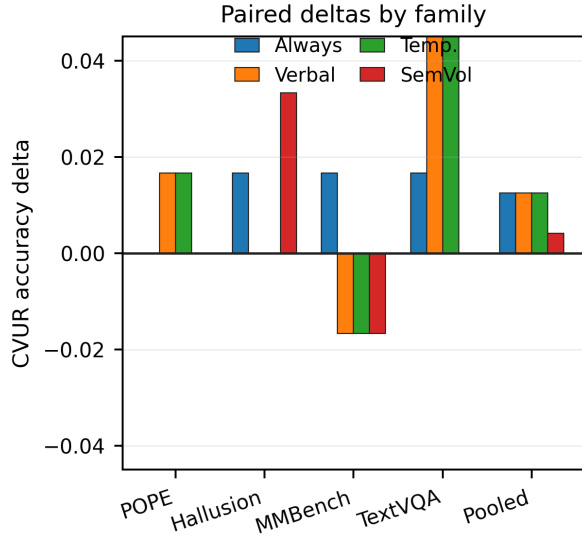


Figure 3: **Family-level paired deltas are mixed despite the 0.004167 pooled delta against the strongest baseline.** The plot motivates conservative, non-uniform-win language.

component is indispensable in every benchmark family.

The failure taxonomy separates trusted-correct, trusted-incorrect, routed-correct, routed-incorrect, high-confidence error, and low-confidence-correct outcomes. On the pooled slice, CVUR records 167 trusted-correct cases, 58 trusted-incorrect cases, 6 routed-or-abstained correct cases, and 9 routed-or-abstained incorrect cases. It also reduces high-confidence errors compared with Always Trust in the taxonomy, but not enough to eliminate them. This matters for deployment: a routed answer is not automatically fixed, and a trusted answer can still be wrong.

The main remaining failure mode is overconfident agreement. If all views inherit the same visual misconception or language prior, consistency can raise confidence in a wrong answer. OCR-oriented checks reduce some text-grounding errors, but they can also fail when the model cannot reliably read the image text. Crop/zoom checks can help when the relevant region is clear, but can be neutral when the benchmark image already presents the target object plainly. These observations explain why the method is best treated as a routing signal to be validated at an operating point, not as a correctness certificate.

The family rows in Table 2 give the main reason for the conservative thesis. The pooled result is positive, but it is not a uniform family-level win. HallusionBench is favorable or tied depending on

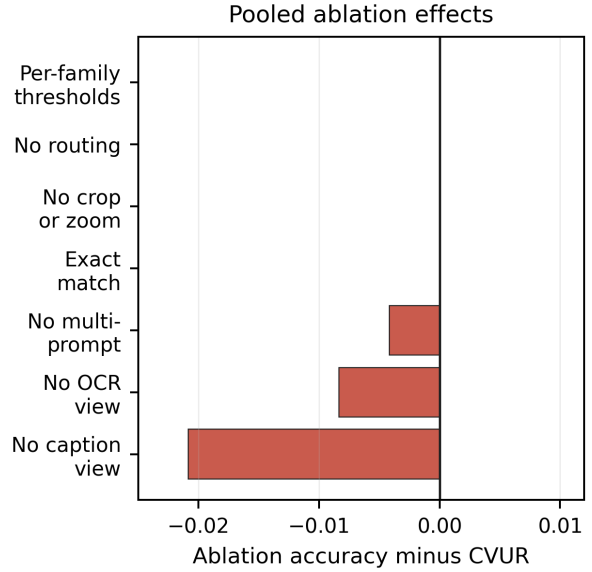


Figure 4: **Ablation effects concentrate in a few signals rather than uniformly across all components.** Matched rows make these diagnostics descriptive, not component-necessity proof.

Table 4: **CVUR operating points vary sharply by family: MMBench English dev has 0.634615 selective risk, while TextVQA validation has 0.111111.** This supports family-specific caution when choosing a threshold.

Family	Coverage	Risk	ECE
All families	0.937500	0.257778	0.194792
HallusionBench	1.000000	0.166667	0.150000
MMBench English dev	0.866667	0.634615	0.504167
POPE	0.983333	0.152542	0.170833
TextVQA validation	0.900000	0.111111	0.104167

the comparator, POPE and TextVQA validation are tied against the semantic-volume baseline, and MMBench English dev is a contrary family. This pattern is consistent with the method design: when the base answerer is broadly uncertain or when multiple views share the same reasoning mistake, view agreement does not necessarily separate correct from incorrect answers.

Table 4 shows that the recorded threshold is not equally appropriate for every family. MMBench English dev combines lower coverage with high selective risk, suggesting that the fixed aggregation is poorly matched to that family. HallusionBench and TextVQA validation are more favorable at the same operating point. This is why the paper reports the pooled threshold as an evaluated protocol choice, not as a universal threshold recommendation.

The full failure taxonomy is reported in Ap-

pendix Table 5, Panel A. It separates two kinds of error that are easy to conflate. Trusted incorrect cases are the direct risk of relying on the answer; CVUR records 58 of them in the pooled slice. Routed incorrect cases are not successes; they are cases where the protocol declined to trust an answer that was still wrong under the benchmark label, and CVUR records 9 of them. A downstream tool or human escalation could in principle repair some routed cases, but this experiment does not measure that second-stage outcome. The main ethical implication is that a trust score should be paired with an actual escalation policy before deployment.

Interpretation for future scaling. The current evidence identifies where a larger study should spend effort. First, threshold selection should be validated separately from test scoring, especially for families like MMBench English dev where the fixed threshold is poorly matched. Second, the caption-evidence signal deserves closer study because removing it gives the largest pooled ablation drop, while other removals tie the full system. Third, a route-to-tool experiment is needed before the method can claim end-to-end utility after abstention. These are not cosmetic extensions; they are the experiments required to turn the present confidence-interface result into a deployment claim.

How to read the routing decision. A trusted CVUR answer should be read as an answer whose available views agree enough to pass the recorded threshold, not as an answer whose correctness has been proven. This distinction is important in the mixed family results. In POPE and TextVQA validation, the operating point is comparatively favorable: coverage remains high and selective risk is below the pooled risk. In MMBench English dev, the same threshold leaves many trusted errors, so a deployment using similar broad reasoning tasks would need a different threshold, a stronger downstream checker, or a model-specific recalibration set. The paper therefore treats the threshold as a measured policy point, not as a transferable constant.

The route/abstain side of the decision has the same limitation. Routing is useful only if the next step is actually better than trusting the answer. In this experiment, routed examples are counted as selective-prediction outcomes; they are not repaired by an external OCR tool, visual cropper, or human annotator. This is why the main text reports

6 routed-or-abstained correct cases and 9 routed-or-abstained incorrect cases instead of converting routed examples into assumed wins. The result supports confidence-aware triage, but the utility of triage must be measured in a follow-up protocol that includes the routed action.

5 Conclusion

CVUR shows that a training-free combination of prompt, crop/zoom, caption-evidence, and OCR agreement can produce a useful confidence score for multimodal selective prediction. The current four-family evidence supports a bounded calibration and routing claim: CVUR has a small pooled accuracy edge, lower ECE-10 than the tested confidence baselines, and an explicit trust-versus-route operating point. It does not support broad accuracy-superiority or component-necessity claims. Future work should evaluate larger official splits and route-to-tool utility after abstention.

Limitations and Ethical Considerations

Limitations. The evaluation is a completed 60-row-per-family slice across four public benchmark families, not full official benchmark coverage. The strongest-baseline accuracy comparison is non-significant, and the family-level paired results are mixed. Route/abstain is evaluated as selective prediction behavior; this paper does not measure whether a downstream tool or human route produces a better final answer. The ablation evidence is also mixed, so individual signal necessity remains unresolved. Finally, the evaluated model/backend is a single multimodal answerer configuration, and additional models may shift the calibration and coverage-risk behavior.

Ethical considerations. Confidence and abstention mechanisms can make systems appear safer than they are. CVUR should not be used as a safety guarantee, especially in medical, legal, or other high-stakes visual decision settings. A routed or abstained example still needs a reliable escalation path, and a trusted example can remain wrong when all views share the same error. Benchmark slices may also underrepresent real-world image quality, accessibility, language, and text-recognition failures.

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Yu Zhang, Chuyang Sun, Kehai Chen, Xuefeng Bai, Yang Xiang, and Min Zhang. 2026b. [Mitigating multimodal hallucination via phase-wise self-reward.](#) *arXiv preprint*.

Reproducibility appendix. The evaluation uses Qwen/Qwen2.5-VL-7B-Instruct with deterministic decoding, a 64-token answer budget, four public benchmark families, and matched baseline/method rows for each selected task.

In this appendix, the evaluated backend denotes that model and decoding setting, and the benchmark bundle denotes the four 60-row public-family slice.

Reproduction requires the selected task identifiers, model outputs, confidence scores, thresholded trust decisions, and the scripts that derive aggregate metrics and plots from those scored rows. The conceptual method figure is a static overview of the protocol and is not used for scoring.

Additional evidence notes. The completed four-family evidence supports the bounded selective-prediction claim, but it does not support broad accuracy superiority or component-necessity claims. The manuscript therefore uses conservative language in the abstract, results, and conclusion.

The reproducibility package separates scored examples from derived summaries. The scored rows contain the task identifiers, benchmark family, method condition, confidence score, trust decision, and correctness label. Derived summaries compute the tables and figures from those rows.

This separation matters because several quantities are easy to misstate if copied by hand. The strongest-baseline delta is 0.004167 in the paired table, while the rounded pooled accuracy difference in the report is 0.004166. Both refer to the same small effect, and neither licenses significance language.

The benchmark bundle should be read as a stress test rather than a claim of benchmark coverage. POPE emphasizes object-presence false trust. HallusionBench emphasizes visual illusion and language-visual conflict. MMBench English dev stresses general multimodal reasoning, and TextVQA validation stresses OCR-sensitive answers.

A larger study would need official full-split runs, additional backends, and route-to-tool evaluation before claiming deployment utility. The present paper is therefore intentionally conservative. It reports that cross-view consistency can produce a useful confidence score, not that it solves multimodal uncertainty.

The appendix tables make the scope limitations reproducible. Table 5 shows that trusted wrong

Table 5: Appendix diagnostics show 58 trusted incorrect cases, 9 routed incorrect cases, and a mixed strongest-baseline family comparison. Panel A gives the pooled failure taxonomy; Panel B gives the family-level comparison against the semantic-volume baseline.

Item	Count/CVUR	Rate/base	Δ	W/L/T
<i>Panel A: pooled failure taxonomy</i>				
Trusted correct	167	0.695833	—	—
Trusted incorrect	58	0.241667	—	—
Routed or abstained correct	6	0.025000	—	—
Routed or abstained incorrect	9	0.037500	—	—
High-confidence error	40	0.166667	—	—
Low-confidence correct	6	0.025000	—	—
<i>Panel B: family comparison against semantic-volume baseline</i>				
HallusionBench	0.833333	0.800000	0.033333	2/0/58
MMBench English dev	0.350000	0.366667	-0.016667	0/1/59
POPE	0.850000	0.850000	0.000000	0/0/60
TextVQA validation	0.850000	0.850000	0.000000	0/0/60

answers remain common enough to matter and records the strongest-baseline family comparisons. Table 6 bounds benchmark provenance.

A reader can replay the main claims by recomputing aggregate accuracy, ECE-10, Brier score, selective risk, paired sign tests, and ablation deltas from the scored rows. The replay should then check that the manuscript does not state a stronger conclusion than those quantities support.

The appendix also records what would be required for a stronger version of the paper. Full official benchmark submissions would replace the 60-row-per-family slice. Additional multimodal backends would test whether the confidence score is model-specific.

A route-to-tool condition would measure whether abstained examples become correct after escalation. These extensions are not assumed in the current evidence. This is why the main manuscript describes CVUR as a confidence-routing interface rather than an end-to-end multimodal reliability solution.

Replay fields. The scored-row replay interface is intentionally narrow. Each evaluated row needs the public benchmark family, a stable task identifier, the method or baseline condition, the deterministic model answer, the correctness label, the confidence score used by that condition, and the resulting trust decision. Those fields are sufficient to recompute pooled accuracy, family-level accuracy, ECE-10, Brier score, coverage, selective risk, route/abstain rate, paired sign tests, and the ablation deltas reported in the paper.

They also prevent a common reproducibility mistake: changing a routed example into a cor-

rected final answer without running and reporting an actual second-stage route-to-tool condition. In this manuscript, routed examples remain selective-prediction events.

Calibration replay. ECE-10 is computed from the confidence scores attached to the trust decision, not from verbal justifications in the generated answer. The Brier score uses the same confidence values against binary correctness labels. Selective risk is computed only over trusted examples at the recorded threshold, while coverage is the trusted fraction of the evaluated slice. These definitions are repeated here because small changes in denominator can make the operating point look stronger than it is. For example, counting routed-correct cases as corrected answers would turn an abstention experiment into an unmeasured tool-use experiment; the paper avoids that conversion.

Paired-test replay. The paired sign tests compare matched task outcomes between CVUR and each baseline. Ties are retained in the descriptive W/L/T counts but omitted from the two-sided sign-test numerator and denominator. This is why the strongest baseline comparison can show a positive pooled delta while still giving $p = 1.000000$: only three matched tasks are non-ties against the semantic-volume baseline. The reported conclusion follows the paired test rather than the decimal ordering alone.

Ablation replay. The ablation rows reuse the same benchmark slice and remove or simplify one signal at a time. Their purpose is diagnostic. The current evidence suggests that caption-style evidence is the most consequential named signal in this slice, but several other removals tie the full method. A larger ablation study would need more tasks and additional model backends before treating any one view as necessary. The main paper therefore uses the ablation table to explain where the confidence score appears to come from, not to claim a causal component proof.

Family-specific replay checks. A verifier should recompute each benchmark family before recomputing the pooled summary. This ordering catches two failure modes. First, the pooled score can look stable while one family carries most of the positive movement. Second, calibration and selective risk can point in different directions across families. In the current evidence, MMBench English dev is the clearest warning case: the recorded operating point

has high selective risk and lower accuracy than several confidence baselines. TextVQA validation and POPE are more favorable at the same threshold. A replay that reports only the pooled row would therefore hide the main reason the manuscript avoids a uniform-win claim.

Threshold replay checks. The recorded threshold should be replayed exactly before any threshold sweep is interpreted. A sweep is useful for drawing the coverage-risk curve, but it must not replace the stated operating point after test outcomes are known. The paper uses the recorded threshold to report 0.937500 coverage and 0.257778 selective risk. A deployment could choose a stricter point, but that would be a new policy choice requiring validation. The appendix states this because selective prediction papers are especially vulnerable to hindsight threshold selection: the same confidence scores can look substantially safer if the threshold is chosen after reading test labels.

Route-action boundary. The route/abstain decision is an interface, not a completed repair. The scored rows record whether CVUR trusted or routed the candidate answer, but they do not include a second-stage OCR tool, human review, search system, or visual cropper that answers routed cases. This boundary is central to the ethics and limitations sections. If a future experiment adds a route action, the replay must add new columns for the routed action, its output, its correctness label, and its budget. Without those columns, a routed case should not be counted as a corrected answer.

Model-backend boundary. All reported rows use one evaluated multimodal answerer configuration. The paper names that backend and decoding budget in the abstract, setup, and main table so that the confidence behavior is not mistaken for a model-independent fact. A stronger study would rerun the protocol on additional open-weight and hosted multimodal backends, then report whether the same threshold and signal weights transfer. The current paper does not assume such transfer. This is why the conclusion says the protocol can produce a useful confidence score on the completed slice, rather than claiming that cross-view consistency is calibrated for all multimodal LLMs.

Citation and benchmark boundary. The bibliography and benchmark provenance serve different roles. Citations ground the motivation for

Table 6: **The benchmark slice uses four public source families with 60 evaluated tasks each.** The table records the paper-facing provenance used to bound the claims and avoid presenting the slice as a full official benchmark submission.

Family	Evaluation role and claim boundary	Tasks
POPE	Object hallucination and absent-object false trust; public slice, not leaderboard coverage	60
HallusionBench	Visual illusion and language-visual conflict; public slice, not exhaustive suite coverage	60
MMBench English dev	General multimodal reasoning and calibration stress; family-level results include underperformance	60
TextVQA validation	OCR-sensitive VQA; route utility measured only as selective prediction	60

calibration, selective prediction, multimodal hallucination, semantic uncertainty, and OCR-sensitive VQA. Benchmark provenance records which public task families were actually scored. A paper-facing claim needs both: literature can motivate a failure mode, but only scored rows can support the local numeric result. Conversely, a benchmark row can support a number without proving novelty or generality. The manuscript keeps these roles separate so that the reader can distinguish prior-work motivation from experiment evidence.